Battleship

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Chapter 1

Battleship

1.1 Projekto nariai

- Justas Kurtinaitis (builder, designer)
- Edgar Dainarovič (architect, critic)
- Kamil Bužinski (facilitator, exhibitor).

1.2 Aprašymas

Šio projekto tikslas - sukurti Battleship žaidimo kopiją.

1.3 Projekto struktūra

1.3.1 1. Projekto logika

1.3.2 2. Vartotojo pasiekiamos funkcijos

1.4 Naudojamos technologijos

Grafiniam interfeisui:

- `Raylib`

Bendradarbiavimui:

- `GitHub`
- `Discord`

1.5 Darbo apimtis

Darbas buvo paskirstytas polygiai.

Chapter 2

Directory Hierarchy

2.1 Directories

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

Chapter 6

Directory Documentation

6.1 include Directory Reference

Files

- file AI.h

  Defines the AI class, which represents a computer-controlled player in a game of Battleship.
- file Board.h

  Defines the Board class, which represents the game board in a game of Battleship.
- file Coordinates.h

  Defines the Coordinates struct, which represents a point in 2D space with x and y coordinates.
- file Game.h

  Defines The game class, which allows users to interact with the game.
- file Human.h

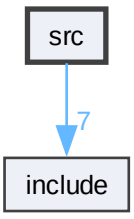
  Defines the Human class, which represents a human player in a game of Battleship.
- file Player.h

  Defines the abstract Player class, which serves as the base class for Human and AI players in a game of Battleship.
- file Ship.h

  Defines the Ship class, which represents a ship in a game of Battleship.

6.2 src Directory Reference

Directory dependency graph for src:



Files

- file AI.cpp

  Implements the AI class.
- file Board.cpp

  Implements the Board class, which represents the game board in a game of Battleship.
- file Game.cpp

  Implementation of Game class.
- file Human.cpp

  Implements the Human class.
- file main.cpp

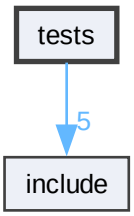
  main game file that runs the game.
- file Player.cpp

  Implements the Player class, the abstract base for Human and AI players in a game of Battleship.
- file Ship.cpp

  Implements the Ship class, which represents a ship in a game of Battleship.

6.3 tests Directory Reference

Directory dependency graph for tests:



Files

- file AITest.cpp
- file AITest.h
- file BoardTest.cpp
- file BoardTest.h
- file HumanTest.cpp
- file HumanTest.h
- file ShipTest.cpp
- file ShipTest.h
- file test.cpp

Chapter 7

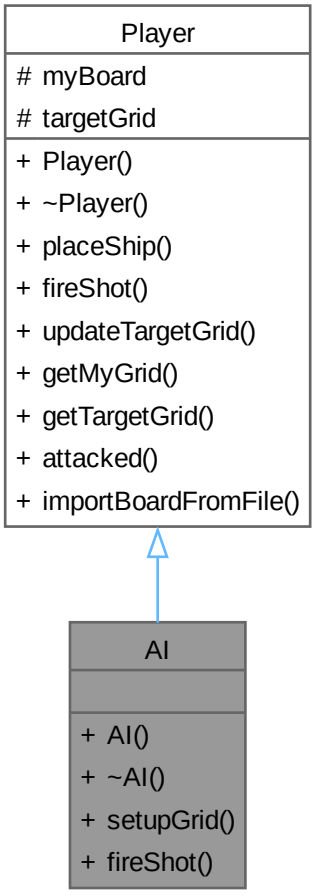
Class Documentation

7.1 AI Class Reference

Represents a computer-controlled player in a game of Battleship.

`#include <AI.h>`

Inheritance diagram for AI:



Collaboration diagram for AI:

Public Member Functions

- AI ()

  Default constructor. Delegates initialisation to Player().
- ~AI ()=default

  Default destructor.
- void setupGrid ()

Automatically places all required ships on the AI's own board.

- Coordinates fireShot () override

  Automatically selects a target cell on the opponent's board.

- bool placeShip (Ship ship)

  Attempts to place a ship on the player's own board.

- void updateTargetGrid (Coordinates coords, bool hit)

  Records the outcome of a shot on the target grid.

- std::vector< std::vector< int > > getMyGrid () const

  Returns a pointer to the player's own 10×10 board grid.

- std::vector< std::vector< int > > getTargetGrid () const

  Returns a pointer to the player's 10×10 target-tracking grid.

- bool attacked (Coordinates attack)

  a wrapper for Board::attack, to be called if atacking this player

- void importBoardFromFile (const std::string &filename)

  Imports the player's board configuration from a file.

Protected Attributes

- Board myBoard
- std::vector< std::vector< int > > targetGrid

7.1.1 Detailed Description

Represents a computer-controlled player in a game of Battleship.

Extends Player with:

- setupGrid() : automatically places all ships on the AI's own board using random-but-valid positions.
- fireShot() : automatically selects an un-fired target cell. The base implementation uses a random strategy; the game logic layer may call more sophisticated selection.

7.1.2 Constructor & Destructor Documentation

7.1.2.1 AI()

```
AI::AI ()
```

Default constructor. Delegates initialisation to Player().

7.1.2.2 ~AI()

```
AI::~AI ()  [default]
```

Default destructor.

7.1.3 Member Function Documentation

7.1.3.1 attacked()

```
bool Player::attacked (
            Coordinates attack)  [inherited]
```

a wrapper for Board::attack, to be called if atacking this player

wrapper for board::attack

Parameters

attack	coordinates to be attacked

Returns

true
false

Parameters

attack	The coordinates of the attack.

Returns

true if the attack hits a ship, false if it misses.

7.1.3.2 fireShot()

```
Coordinates AI::fireShot ()  [override], [virtual]
```

Automatically selects a target cell on the opponent's board.

Picks a cell whose targetGrid value is 0 (not yet attacked). The selection strategy (random vs. hunt/target) is determined by the logic implemented in the body of this method.

Returns

Coordinates The cell the AI chooses to attack.

Implements Player.

7.1.3.3 getMyGrid()

```
std::vector< std::vector< int > > Player::getMyGrid () const  [inherited]
```

Returns a pointer to the player's own 10×10 board grid.

Grid encoding: 0 = empty, 1 = ship, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D array of myBoard.

7.1.3.4 getTargetGrid()

```
std::vector< std::vector< int > > Player::getTargetGrid () const  [inherited]
```

Returns a pointer to the player's 10×10 target-tracking grid.

Grid encoding: 0 = unknown, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D target grid array.

7.1.3.5 importBoardFromFile()

```
void Player::importBoardFromFile (
            const std::string & filename)  [inline], [inherited]
```

Imports the player's board configuration from a file.

Parameters

filename	The name of the file containing the board configuration (ship placements).

7.1.3.6 placeShip()

```
bool Player::placeShip (
            Ship ship)  [inherited]
```

Attempts to place a ship on the player's own board.

Delegates to Board::placeShip, which validates bounds and overlap.

Parameters

ship	The ship to place (start/end coordinates must already be set).

Returns

true if the ship was placed successfully.

false if the placement is invalid (out of bounds or overlapping).

7.1.3.7 setupGrid()

```
void AI::setupGrid ()
```

Automatically places all required ships on the AI's own board.

Generates random start coordinates and orientation for each ship, retrying until Board::placeShip() accepts the placement. Called once during the Setup phase (see activity diagram).

7.1.3.8 updateTargetGrid()

```
void Player::updateTargetGrid (
            Coordinates coords,
            bool hit)  [inherited]
```

Records the outcome of a shot on the target grid.

Should be called after the opponent's Board::attack() returns so that the player's local view stays in sync with the opponent's board state.

Parameters

coords	The cell that was attacked.
hit	true if the shot hit a ship, false if it missed.

7.1.4 Member Data Documentation

7.1.4.1 myBoard

```
Board Player::myBoard  [protected], [inherited]
```

The player's own board, containing their ships.

7.1.4.2 targetGrid

```
std::vector<std::vector<int> > Player::targetGrid  [protected], [inherited]
```

Local tracking grid for shots fired at the opponent. 0 = unknown, 2 = hit, 3 = miss (mirrors Board grid encoding).

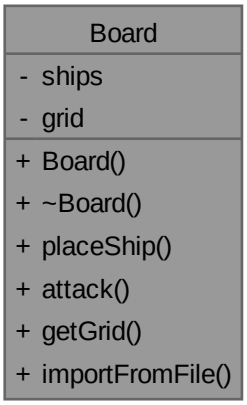
The documentation for this class was generated from the following files:

- include/AI.h
- src/AI.cpp

7.2 Board Class Reference

```
#include <Board.h>
```

Collaboration diagram for Board:



Public Member Functions

- Board ()

  Default constructor that initializes the board with default values.
- ~Board ()=default

  Places a ship on the board at the specified coordinates.
- bool placeShip (Ship ship)
- bool attack (Coordinates attack)

  Attacks the specified coordinates on the board.
- std::vector< std::vector< int > > getGrid () const

  Returns a pointer to the 2D array representing the game board.
- void importFromFile (const std::string &filename)

  Imports the game board configuration from a file (file format - coordinates of the ships).

Private Attributes

- std::vector< Ship > ships
- std::vector< std::vector< int > > grid

7.2.1 Constructor & Destructor Documentation

7.2.1.1 Board()

```
Board::Board ()
```

Default constructor that initializes the board with default values.

7.2.1.2 ~Board()

```
Board::~Board ()  [default]
```

Places a ship on the board at the specified coordinates.

Parameters

ship	The ship to be placed on the board.

Returns

true if the ship was successfully placed, false if the placement is invalid (e.g., out of bounds or overlapping with another ship).

Destructor for the Board class. arrays and vectors will automatically clean up their memory, so no explicit cleanup is necessary in this destructor.

7.2.2 Member Function Documentation

7.2.2.1 attack()

```
bool Board::attack (
            Coordinates attack)
```

Attacks the specified coordinates on the board.

Parameters

attack	The coordinates of the attack.

Returns

true if the attack hits a ship, false if it misses.

7.2.2.2 getGrid()

```
std::vector< std::vector< int > > Board::getGrid () const  [inline]
```

Returns a pointer to the 2D array representing the game board.

Returns

A pointer to the 2D array representing the game board

7.2.2.3 importFromFile()

```
void Board::importFromFile (
            const std::string & filename)
```

Imports the game board configuration from a file (file format - coordinates of the ships).

Parameters

filename	The name of the file containing the game board configuration.

7.2.2.4 placeShip()

```
bool Board::placeShip (
            Ship ship)
```

7.2.3 Member Data Documentation

7.2.3.1 grid

```
std::vector<std::vector<int> > Board::grid  [private]
```

A 2D array representing the game board, where 0 indicates an empty cell, 1 indicates a cell occupied by a ship, 2 indicates hit, and 3 indicates a miss.

7.2.3.2 ships

```
std::vector<Ship> Board::ships  [private]
```

An array of ships on the board.

The documentation for this class was generated from the following files:

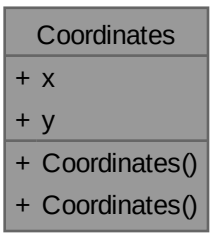
- include/Board.h
- src/Board.cpp

7.3 Coordinates Class Reference

Represents a point in 2D space with x and y coordinates.

`#include <Coordinates.h>`

Collaboration diagram for Coordinates:



Public Member Functions

- Coordinates ()

  Default constructor that initializes the coordinates to (0, 0).
- Coordinates (int x, int y)

  Parameterized constructor that initializes the coordinates to the specified values.

Public Attributes

- int x
- int y

7.3.1 Detailed Description

Represents a point in 2D space with x and y coordinates.

7.3.2 Constructor & Destructor Documentation

7.3.2.1 Coordinates() [1/2]

```
Coordinates::Coordinates ()  [inline]
```

Default constructor that initializes the coordinates to (0, 0).

7.3.2.2 Coordinates() [2/2]

```
Coordinates::Coordinates (
            int x,
            int y)  [inline]
```

Parameterized constructor that initializes the coordinates to the specified values.

Parameters

x	
y	

7.3.3 Member Data Documentation

7.3.3.1 x

```
int Coordinates::x
```

7.3.3.2 y

```
int Coordinates::y
```

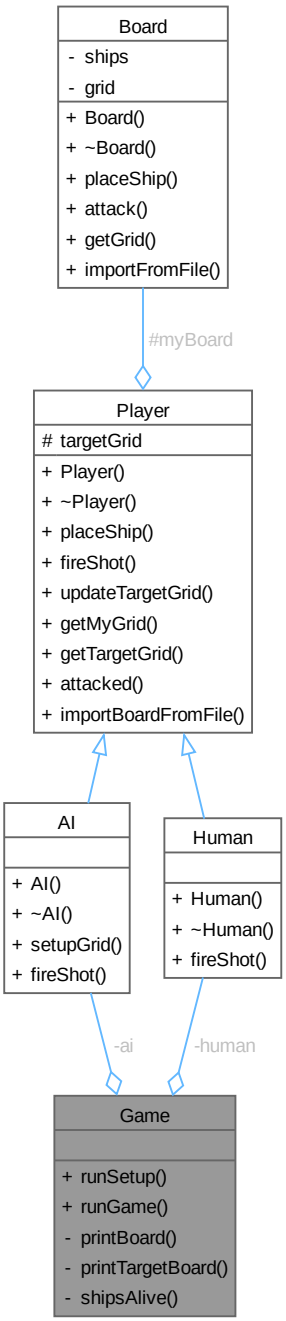
The documentation for this class was generated from the following file:

- include/Coordinates.h

7.4 Game Class Reference

```
#include <Game.h>
```

Collaboration diagram for Game:



Public Member Functions

- void runSetup ()

  runs thru ai and human board setup
- void runGame ()

  runs the game loop

Private Member Functions

- void printBoard (std::vector< std::vector< int > > board)

  prints out the given board parameter
- void printTargetBoard (std::vector< std::vector< int > > board)

  checks if there are alive ships on a given board
- bool shipsAlive (std::vector< std::vector< int > > board)

  checks if there are alive ships on a given board

Private Attributes

- AI ai

  The AI player.
- Human human

  The human player.

7.4.1 Member Function Documentation

7.4.1.1 printBoard()

```
void Game::printBoard (
            std::vector< std::vector< int > > board)  [private]
```

prints out the given board parameter

Parameters

board	Players board

7.4.1.2 printTargetBoard()

```
void Game::printTargetBoard (
            std::vector< std::vector< int > > board)  [private]
```

checks if there are alive ships on a given board

Parameters

board	Players board

Returns

true there are alive ships

false no alive ships are left

prints out the given board parameter, but hides the ships

Parameters

board	

7.4.1.3 runGame()

```
void Game::runGame ()
```

runs the game loop

7.4.1.4 runSetup()

```
void Game::runSetup ()
```

runs thru ai and human board setup

7.4.1.5 shipsAlive()

```
bool Game::shipsAlive (
            std::vector< std::vector< int > > board)  [private]
```

checks if there are alive ships on a given board

Parameters

board	

Returns

true there are alive ships, else false

7.4.2 Member Data Documentation

7.4.2.1 ai

```
AI Game::ai  [private]
```

The AI player.

7.4.2.2 human

```
Human Game::human  [private]
```

The human player.

The documentation for this class was generated from the following files:

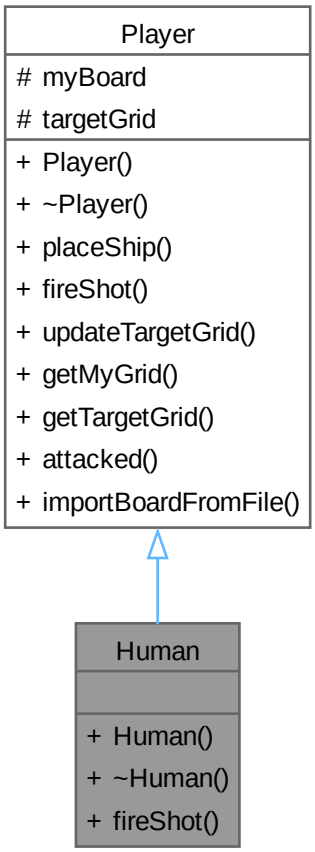
- include/Game.h
- src/Game.cpp

7.5 Human Class Reference

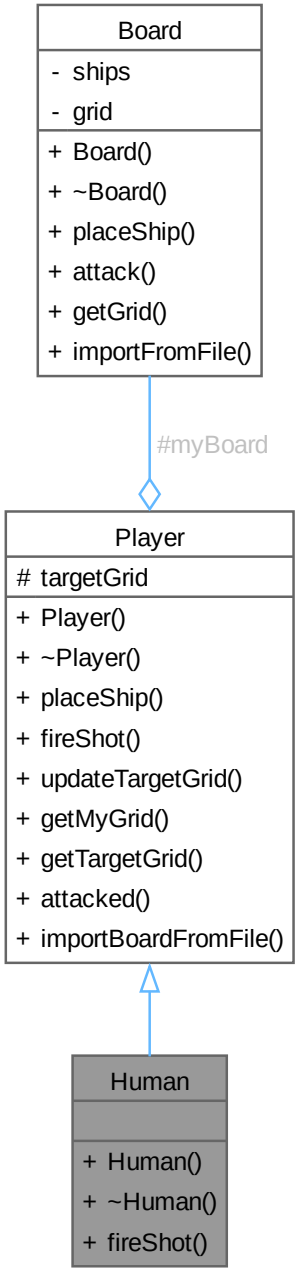
Represents a human-controlled player in a game of Battleship.

```
#include <Human.h>
```

Inheritance diagram for Human:



Collaboration diagram for Human:



Public Member Functions

- Human ()

  Default constructor. Delegates initialisation to Player().
- ~Human ()=default

  Default destructor.
- Coordinates fireShot () override

Prompts the human player to enter attack coordinates.

- bool placeShip (Ship ship)

  Attempts to place a ship on the player's own board.
- void updateTargetGrid (Coordinates coords, bool hit)

  Records the outcome of a shot on the target grid.
- std::vector< std::vector< int > > getMyGrid () const

  Returns a pointer to the player's own 10×10 board grid.
- std::vector< std::vector< int > > getTargetGrid () const

  Returns a pointer to the player's 10×10 target-tracking grid.
- bool attacked (Coordinates attack)

  a wrapper for Board::attack, to be called if atacking this player
- void importBoardFromFile (const std::string &filename)

  Imports the player's board configuration from a file.

Protected Attributes

- Board myBoard
- std::vector< std::vector< int > > targetGrid

7.5.1 Detailed Description

Represents a human-controlled player in a game of Battleship.

Extends Player with a fireShot() implementation that obtains the target coordinates from user input (keyboard). Ship placement is interactive: the game loop is responsible for reading the desired position and rotation from the user and calling placeShip() accordingly.

7.5.2 Constructor & Destructor Documentation

7.5.2.1 Human()

```
Human::Human ()
```

Default constructor. Delegates initialisation to Player().

7.5.2.2 ~Human()

```
Human::~Human ()  [default]
```

Default destructor.

7.5.3 Member Function Documentation

7.5.3.1 attacked()

```
bool Player::attacked (
            Coordinates attack)  [inherited]
```

a wrapper for Board::attack, to be called if atacking this player

wrapper for board::attack

Parameters

attack	coordinates to be attacked

Returns

true
false

Parameters

attack	The coordinates of the attack.

Returns

true if the attack hits a ship, false if it misses.

7.5.3.2 fireShot()

```
Coordinates Human::fireShot ()  [override], [virtual]
```

Prompts the human player to enter attack coordinates.

Reads a target cell from standard input, validates that the cell has not already been fired upon (targetGrid value is 0), and returns the chosen coordinates. Re-prompts on invalid input.

Returns

Coordinates The cell the human player wishes to attack.

Implements Player.

7.5.3.3 getMyGrid()

```
std::vector< std::vector< int > > Player::getMyGrid () const  [inherited]
```

Returns a pointer to the player's own 10×10 board grid.

Grid encoding: 0 = empty, 1 = ship, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D array of myBoard.

7.5.3.4 getTargetGrid()

```
std::vector< std::vector< int > > Player::getTargetGrid () const  [inherited]
```

Returns a pointer to the player's 10×10 target-tracking grid.

Grid encoding: 0 = unknown, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D target grid array.

7.5.3.5 importBoardFromFile()

```
void Player::importBoardFromFile (
            const std::string & filename)  [inline], [inherited]
```

Imports the player's board configuration from a file.

Parameters

filename	The name of the file containing the board configuration (ship placements).

7.5.3.6 placeShip()

```
bool Player::placeShip (
            Ship ship)  [inherited]
```

Attempts to place a ship on the player's own board.

Delegates to Board::placeShip, which validates bounds and overlap.

Parameters

ship	The ship to place (start/end coordinates must already be set).

Returns

true if the ship was placed successfully.

false if the placement is invalid (out of bounds or overlapping).

7.5.3.7 updateTargetGrid()

```
void Player::updateTargetGrid (
            Coordinates coords,
            bool hit)  [inherited]
```

Records the outcome of a shot on the target grid.

Should be called after the opponent's Board::attack() returns so that the player's local view stays in sync with the opponent's board state.

Parameters

coords	The cell that was attacked.
hit	true if the shot hit a ship, false if it missed.

7.5.4 Member Data Documentation

7.5.4.1 myBoard

```
Board Player::myBoard  [protected], [inherited]
```

The player's own board, containing their ships.

7.5.4.2 targetGrid

```
std::vector<std::vector<int> > Player::targetGrid  [protected], [inherited]
```

Local tracking grid for shots fired at the opponent. 0 = unknown, 2 = hit, 3 = miss (mirrors Board grid encoding).

The documentation for this class was generated from the following files:

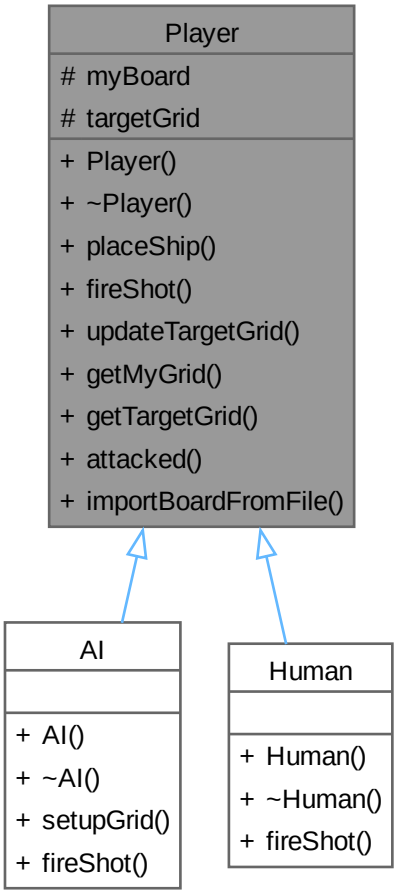
- include/Human.h
- src/Human.cpp

7.6 Player Class Reference

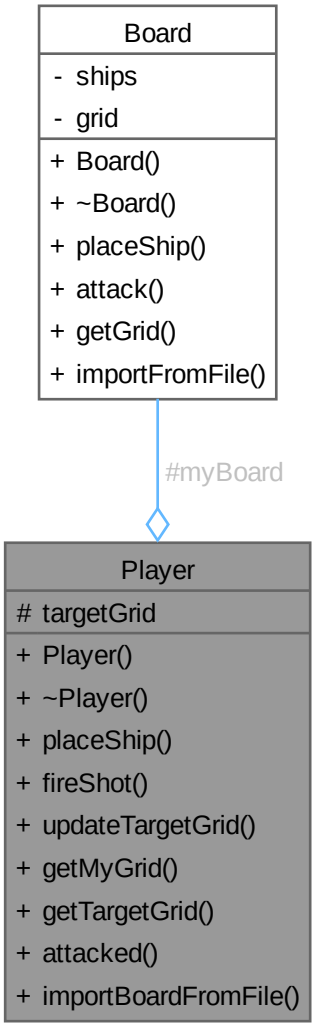
Abstract base class representing a player in a game of Battleship.

```
#include <Player.h>
```

Inheritance diagram for Player:



Collaboration diagram for Player:



Public Member Functions

- Player ()

  Default constructor. Initialises the target grid to all zeros (no shots fired yet).
- virtual ~Player ()=default

  Virtual destructor. Derived classes clean up their own resources.
- bool placeShip (Ship ship)

  Attempts to place a ship on the player's own board.
- virtual Coordinates fireShot ()=0

  Selects the coordinates for the next attack on the opponent's board.
- void updateTargetGrid (Coordinates coords, bool hit)

  Records the outcome of a shot on the target grid.
- std::vector< std::vector< int > > getMyGrid () const

Returns a pointer to the player's own 10×10 board grid.

- std::vector< std::vector< int > > getTargetGrid () const

  Returns a pointer to the player's 10×10 target-tracking grid.

- bool attacked (Coordinates attack)

  a wrapper for Board::attack, to be called if atacking this player

- void importBoardFromFile (const std::string &filename)

  Imports the player's board configuration from a file.

Protected Attributes

- Board myBoard
- std::vector< std::vector< int > > targetGrid

7.6.1 Detailed Description

Abstract base class representing a player in a game of Battleship.

Holds the player's own board (where their ships reside) and a target grid (a local record of shots fired at the opponent's board). Concrete subclasses must implement fireShot(), which differs between a Human (user input) and an AI (automated selection).

7.6.2 Constructor & Destructor Documentation

7.6.2.1 Player()

```
Player::Player ()
```

Default constructor. Initialises the target grid to all zeros (no shots fired yet).

7.6.2.2 ~Player()

```
virtual Player::~Player ()  [virtual], [default]
```

Virtual destructor. Derived classes clean up their own resources.

7.6.3 Member Function Documentation

7.6.3.1 attacked()

```
bool Player::attacked (
            Coordinates attack)
```

a wrapper for Board::attack, to be called if atacking this player

wrapper for board::attack

Parameters

attack	coordinates to be attacked

Returns

true

false

Parameters

attack	The coordinates of the attack.

Returns

true if the attack hits a ship, false if it misses.

7.6.3.2 fireShot()

```
virtual Coordinates Player::fireShot ()  [pure virtual]
```

Selects the coordinates for the next attack on the opponent's board.

Pure virtual — Human reads from user input; AI computes a target automatically.

Returns

Coordinates The cell to attack on the opponent's board.

Implemented in AI, and Human.

7.6.3.3 getMyGrid()

```
std::vector< std::vector< int > > Player::getMyGrid () const
```

Returns a pointer to the player's own 10×10 board grid.

Grid encoding: 0 = empty, 1 = ship, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D array of myBoard.

7.6.3.4 getTargetGrid()

```
std::vector< std::vector< int > > Player::getTargetGrid () const
```

Returns a pointer to the player's 10×10 target-tracking grid.

Grid encoding: 0 = unknown, 2 = hit, 3 = miss.

Returns

vector<vector<int>> internal 2-D target grid array.

7.6.3.5 importBoardFromFile()

```
void Player::importBoardFromFile (
            const std::string & filename)  [inline]
```

Imports the player's board configuration from a file.

Parameters

filename	The name of the file containing the board configuration (ship placements).

7.6.3.6 placeShip()

```
bool Player::placeShip (
            Ship ship)
```

Attempts to place a ship on the player's own board.

Delegates to Board::placeShip, which validates bounds and overlap.

Parameters

ship	The ship to place (start/end coordinates must already be set).

Returns

true if the ship was placed successfully.

false if the placement is invalid (out of bounds or overlapping).

7.6.3.7 updateTargetGrid()

```
void Player::updateTargetGrid (
            Coordinates coords,
            bool hit)
```

Records the outcome of a shot on the target grid.

Should be called after the opponent's Board::attack() returns so that the player's local view stays in sync with the opponent's board state.

Parameters

coords	The cell that was attacked.
hit	true if the shot hit a ship, false if it missed.

7.6.4 Member Data Documentation

7.6.4.1 myBoard

```
Board Player::myBoard  [protected]
```

The player's own board, containing their ships.

7.6.4.2 targetGrid

```
std::vector<std::vector<int> > Player::targetGrid  [protected]
```

Local tracking grid for shots fired at the opponent. 0 = unknown, 2 = hit, 3 = miss (mirrors Board grid encoding).

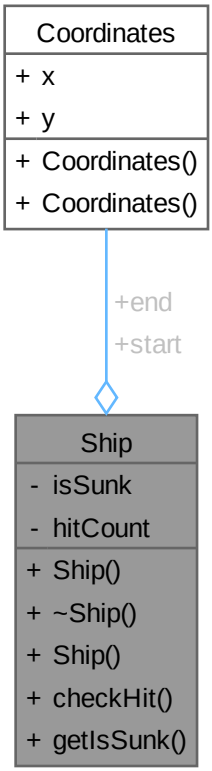
The documentation for this class was generated from the following files:

- include/Player.h
- src/Player.cpp

7.7 Ship Class Reference

```
#include <Ship.h>
```

Collaboration diagram for Ship:



Public Member Functions

- Ship ()

  Default constructor that initializes the ship with default values.
- ~Ship ()=default

  Destructor for the Ship class. Since we are not dynamically allocating any memory in this class, we can use the default destructor provided by the compiler.
- Ship (Coordinates start, Coordinates end)

  Parameterized constructor that initializes the ship with the specified starting and ending coordinates.
- bool checkHit (Coordinates attack)

  Checks if the given attack coordinates hit the ship.
- bool getIsSunk ()

  Returns whether the ship has been sunk.

Public Attributes

- Coordinates start
- Coordinates end

Private Attributes

- bool isSunk
- int hitCount

7.7.1 Constructor & Destructor Documentation

7.7.1.1 Ship() [1/2]

```
Ship::Ship ()  [inline]
```

Default constructor that initializes the ship with default values.

7.7.1.2 ~Ship()

```
Ship::~Ship ()  [default]
```

Destructor for the Ship class. Since we are not dynamically allocating any memory in this class, we can use the default destructor provided by the compiler.

7.7.1.3 Ship() [2/2]

```
Ship::Ship (
            Coordinates start,
            Coordinates end)
```

Parameterized constructor that initializes the ship with the specified starting and ending coordinates.

Parameters

start	
end	

7.7.2 Member Function Documentation

7.7.2.1 checkHit()

```
bool Ship::checkHit (
            Coordinates attack)
```

Checks if the given attack coordinates hit the ship.

Parameters

attack	The coordinates of the attack.

Returns

true if the attack hits the ship, false otherwise.

7.7.2.2 getIsSunk()

```
bool Ship::getIsSunk ()
```

Returns whether the ship has been sunk.

Returns

true if the ship has been sunk, false otherwise.

7.7.3 Member Data Documentation

7.7.3.1 end

```
Coordinates Ship::end
```

The ending coordinates of the ship.

7.7.3.2 hitCount

```
int Ship::hitCount  [private]
```

The number of hits the ship has taken.

7.7.3.3 isSunk

```
bool Ship::isSunk  [private]
```

A flag indicating whether the ship has been sunk.

7.7.3.4 start

```
Coordinates Ship::start
```

The starting coordinates of the ship.

The documentation for this class was generated from the following files:

- include/Ship.h
- src/Ship.cpp

Chapter 8

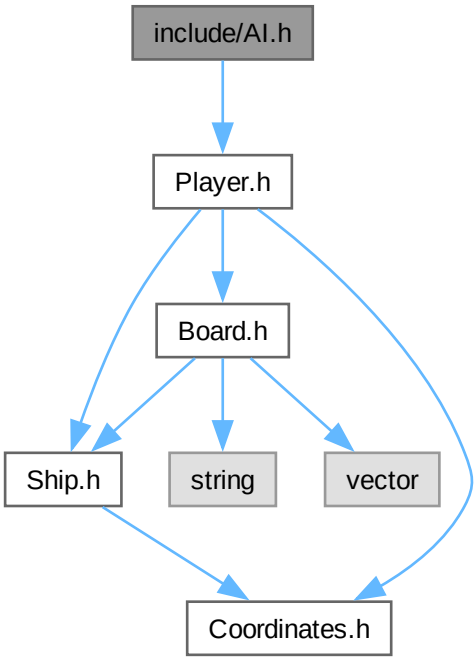
File Documentation

8.1 include/AI.h File Reference

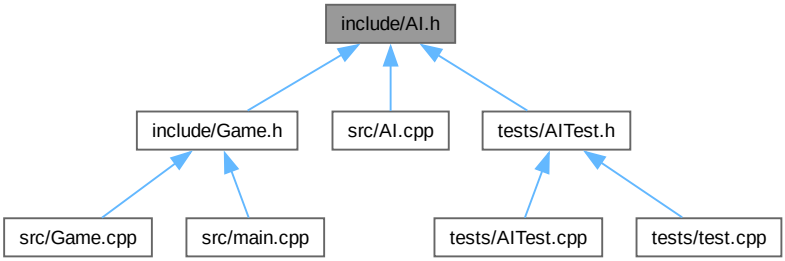
Defines the AI class, which represents a computer-controlled player in a game of Battleship.

```
#include "Player.h"
```

Include dependency graph for AI.h:



This graph shows which files directly or indirectly include this file:



Classes

- class AI

  Represents a computer-controlled player in a game of Battleship.

8.1.1 Detailed Description

Defines the AI class, which represents a computer-controlled player in a game of Battleship.

8.2 AI.h

Go to the documentation of this file.

```
00001
00006
00007 #pragma once
00008
00009 #include "Player.h"
00010
00022 class AI : public Player {
00023 public:
00027   AI();
00028
00032   ~AI() = default;
00033
00041   void setupGrid();
00042
00052   Coordinates fireShot() override;
00053 };
```

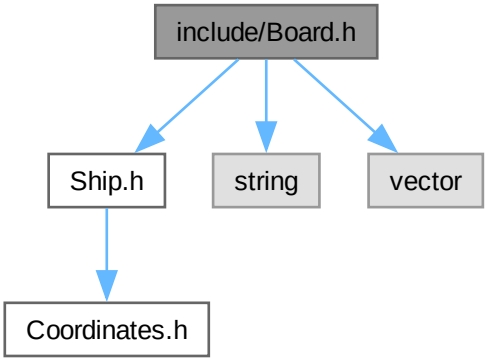
8.3 include/Board.h File Reference

Defines the Board class, which represents the game board in a game of Battleship.

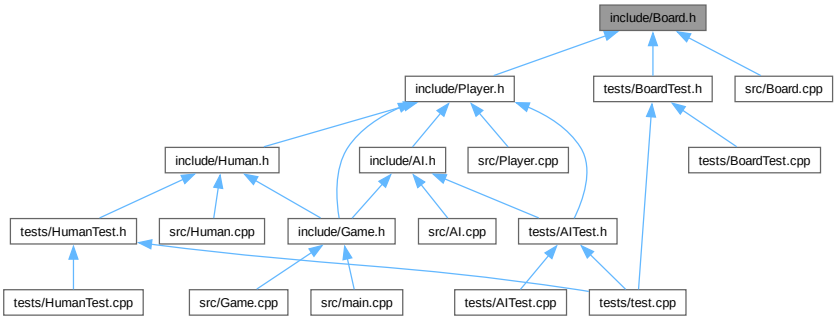
```
#include "Ship.h"
#include <string>
```

```
#include <vector>
```

Include dependency graph for Board.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Board

8.3.1 Detailed Description

Defines the Board class, which represents the game board in a game of Battleship.

8.4 Board.h

Go to the documentation of this file.

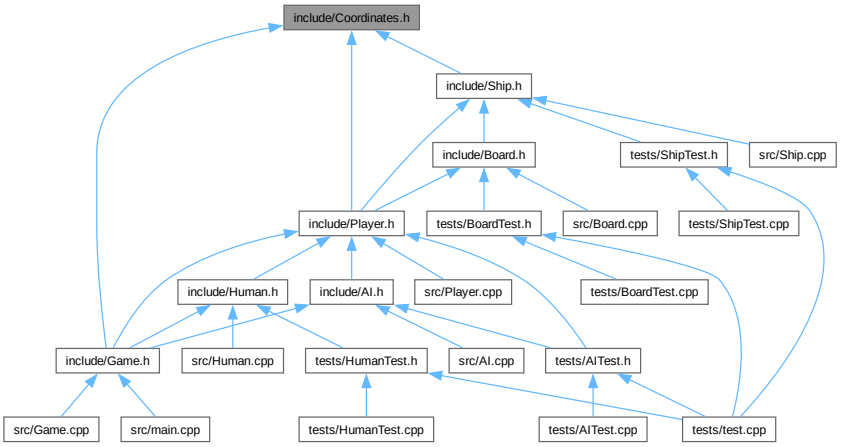
```
00001
```

```
00006
00007 #pragma once
00008
00009 #include "Ship.h"
00010 #include <string>
00011 #include <vector>
00012
00013 /****
00014  * @class Board
00015  * @brief Represents the game board in a game of Battleship.
00016  */
00017 class Board {
00018 private:
00019   std::vector<Ship> ships;
00020   std::vector<std::vector<int»
00021       grid;
00024 public:
00028   Board();
00041   ~Board() = default;
00042   bool placeShip(Ship ship);
00048   bool attack(Coordinates attack);
00049
00055   std::vector<std::vector<int» getGrid() const { return grid; }
00056
00064   void importFromFile(const std::string &filename);
00065 };
```

8.5 include/Coordinates.h File Reference

Defines the Coordinates struct, which represents a point in 2D space with x and y coordinates.

This graph shows which files directly or indirectly include this file:



Classes

- class Coordinates

  Represents a point in 2D space with x and y coordinates.

8.5.1 Detailed Description

Defines the Coordinates struct, which represents a point in 2D space with x and y coordinates.

8.6 Coordinates.h

Go to the documentation of this file.

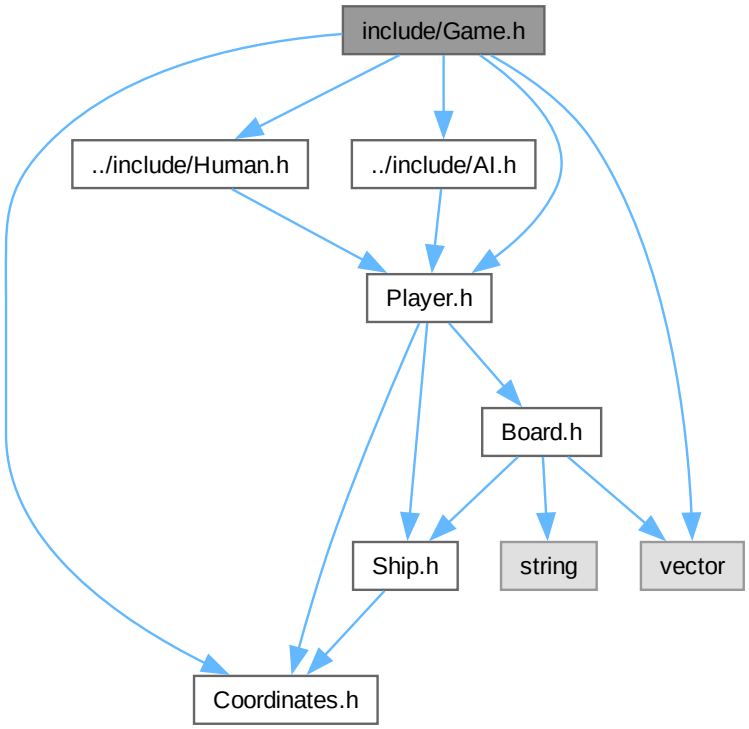
```
00001
00006
00007 #pragma once
00008
00014 struct Coordinates {
00015   int x;
00016   int y;
00017
00021   Coordinates() : x(0), y(0) {}
00029   Coordinates(int x, int y) : x(x), y(y) {}
00030 };
```

8.7 include/Game.h File Reference

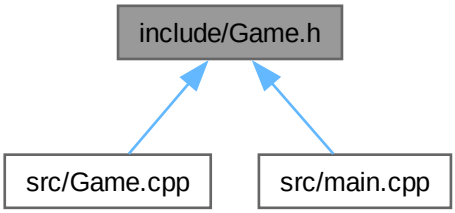
Defines The game class, which allows users to interact with the game.

```
#include "../include/AI.h"
#include "../include/Coordinates.h"
#include "../include/Human.h"
#include "../include/Player.h"
#include <vector>
```

Include dependency graph for Game.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Game

8.7.1 Detailed Description

Defines The game class, which allows users to interact with the game.

8.8 Game.h

Go to the documentation of this file.

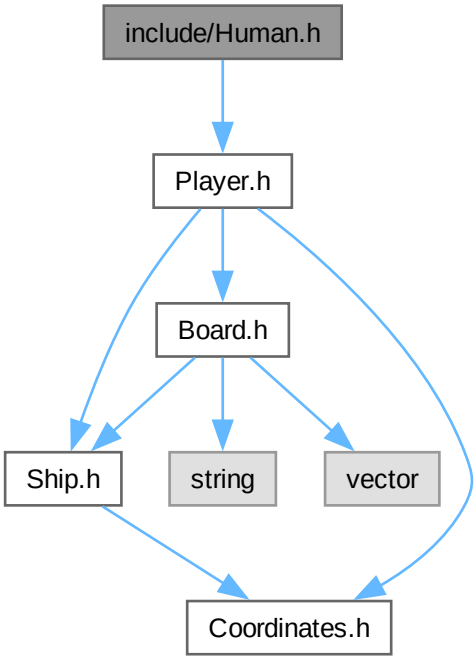
```
00001
00006
00007 #pragma once
00008 #include "../include/AI.h"
00009 #include "../include/Coordinates.h"
00010 #include "../include/Human.h"
00011 #include "../include/Player.h"
00012 #include <vector>
00013 /****
00014  * @class Game
00015  * @brief Implements basic game functions
00016  *
00017  */
00018 class Game {
00019 private:
00025   void printBoard(std::vector<std::vector<int» board);
00033
00039   void printTargetBoard(std::vector<std::vector<int» board);
00046   bool shipsAlive(std::vector<std::vector<int» board);
00047   AI ai;
00048   Human human;
00049
00050 public:
00051   // Game();
00052   // ~Game();
00057   void runSetup();
00062   void runGame();
00063 };
```

8.9 include/Human.h File Reference

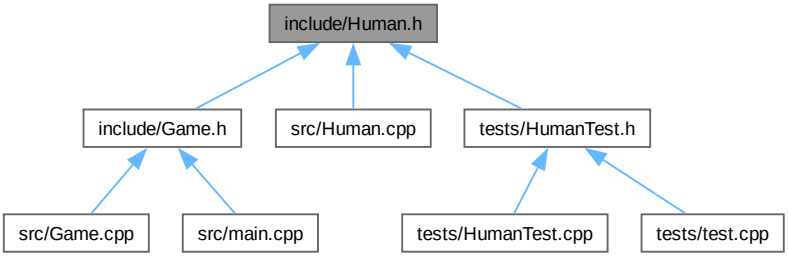
Defines the Human class, which represents a human player in a game of Battleship.

```
#include "Player.h"
```

Include dependency graph for Human.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Human

  Represents a human-controlled player in a game of Battleship.

8.9.1 Detailed Description

Defines the Human class, which represents a human player in a game of Battleship.

8.10 Human.h

Go to the documentation of this file.

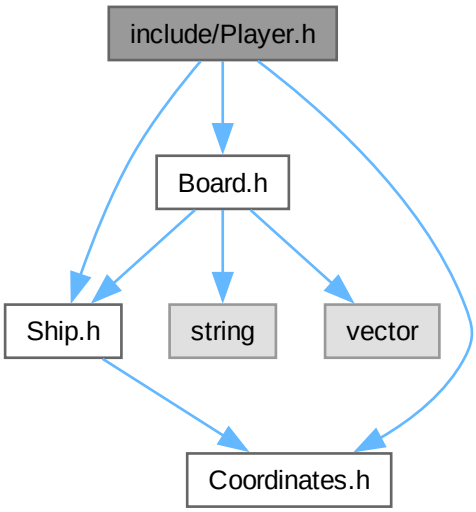
```
00001
00006
00007 #pragma once
00008
00009 #include "Player.h"
00010
00020 class Human : public Player {
00021 public:
00025   Human();
00026
00030   ~Human() = default;
00031
00041   Coordinates fireShot() override;
00042 };
```

8.11 include/Player.h File Reference

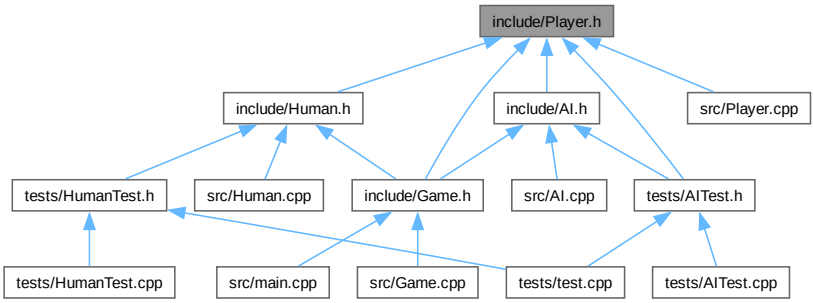
Defines the abstract Player class, which serves as the base class for Human and AI players in a game of Battleship.

```
#include "Board.h"
#include "Coordinates.h"
#include "Ship.h"
```

Include dependency graph for Player.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Player

  Abstract base class representing a player in a game of Battleship.

8.11.1 Detailed Description

Defines the abstract Player class, which serves as the base class for Human and AI players in a game of Battleship.

8.12 Player.h

Go to the documentation of this file.

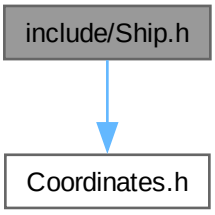
```
00001
00006
00007 #pragma once
00008
00009 #include "Board.h"
00010 #include "Coordinates.h"
00011 #include "Ship.h"
00012
00022 class Player {
00023 protected:
00024   Board myBoard;
00025   std::vector<std::vector<int» targetGrid;
00028
00029 public:
00034   Player();
00035
00039   virtual ~Player() = default;
00040
00050   bool placeShip(Ship ship);
00051
00060   virtual Coordinates fireShot() = 0;
00061
00071   void updateTargetGrid(Coordinates coords, bool hit);
00072
00080   std::vector<std::vector<int» getMyGrid() const;
00081
00089   std::vector<std::vector<int» getTargetGrid() const;
00090
00098   bool attacked(Coordinates attack);
00099
00106   void importBoardFromFile(const std::string &filename) {
00107     myBoard.importFromFile(filename);
00108   }
00109 };
```

8.13 include/Ship.h File Reference

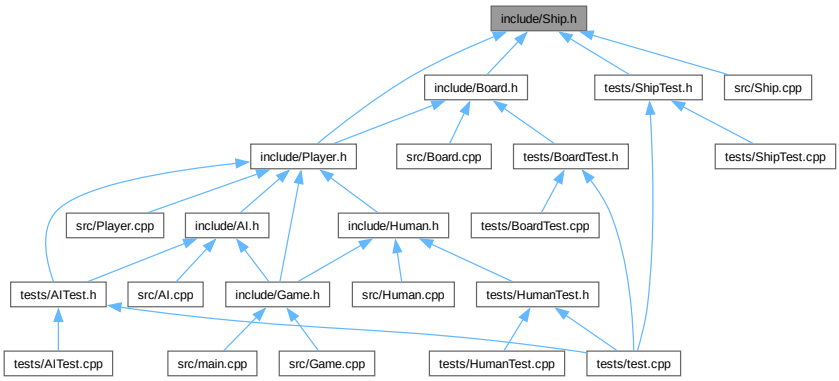
Defines the Ship class, which represents a ship in a game of Battleship.

```
#include "Coordinates.h"
```

Include dependency graph for Ship.h:



This graph shows which files directly or indirectly include this file:



Classes

- class Ship

8.13.1 Detailed Description

Defines the Ship class, which represents a ship in a game of Battleship.

8.14 Ship.h

Go to the documentation of this file.

```
00001
00006
00007 #pragma once
00008
00009 #include "Coordinates.h"
00010
00011 /****
00012  * @class Ship
00013  * @brief Represents a ship in a game of Battleship.
00014  */
00015 class Ship {
00016 private:
00017   bool isSunk;
00018   int hitCount;
00019 public:
00020   Coordinates start;
00021   Coordinates end;
00022
00026   Ship() : isSunk(false), hitCount(0), start(), end() {};
00032   ~Ship() = default;
00040   Ship(Coordinates start, Coordinates end);
00047
00048   bool checkHit(Coordinates attack);
00054   bool getIsSunk();
00055 };
```

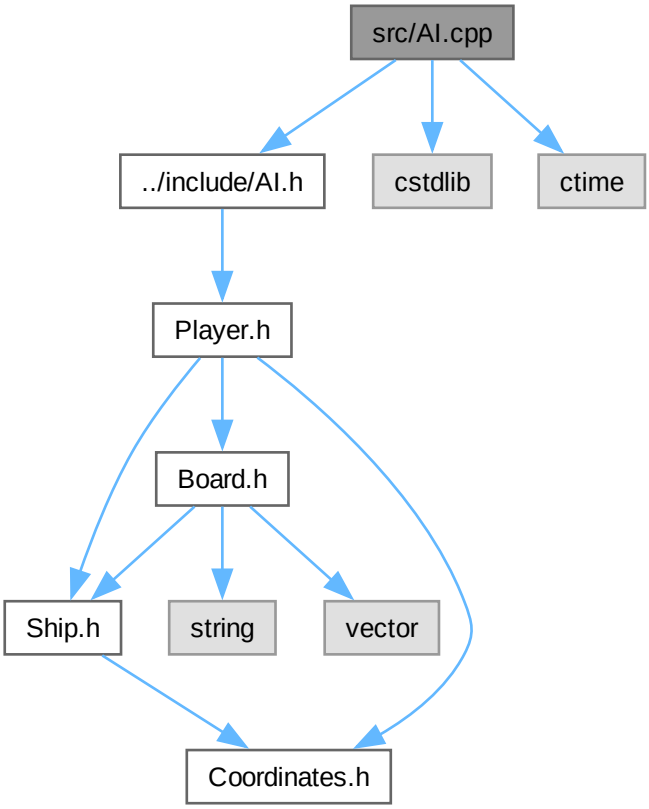
8.15 README.md File Reference

8.16 src/AI.cpp File Reference

Implements the AI class.

```
#include "../include/AI.h"
#include <cstdlib>
#include <ctime>
```

Include dependency graph for AI.cpp:



Variables

- static const int SHIP_SIZES [ ] = {5, 4, 3, 3, 2}
- static const int SHIP_COUNT = 5

8.16.1 Detailed Description

Implements the AI class.

8.16.2 Variable Documentation

8.16.2.1 SHIP_COUNT

```
const int SHIP_COUNT = 5  [static]
```

8.16.2.2 SHIP_SIZES

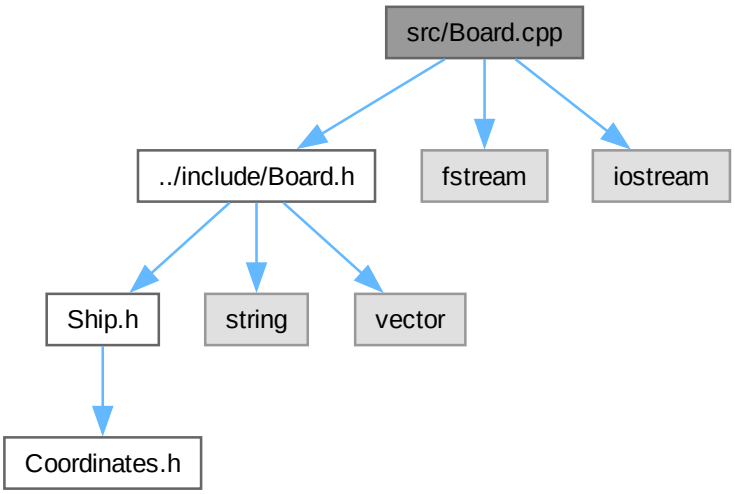
```
const int SHIP_SIZES[] = {5, 4, 3, 3, 2}  [static]
```

8.17 src/Board.cpp File Reference

Implements the Board class, which represents the game board in a game of Battleship.

```
#include "../include/Board.h"
#include <fstream>
#include <iostream>
```

Include dependency graph for Board.cpp:



8.17.1 Detailed Description

Implements the Board class, which represents the game board in a game of Battleship.

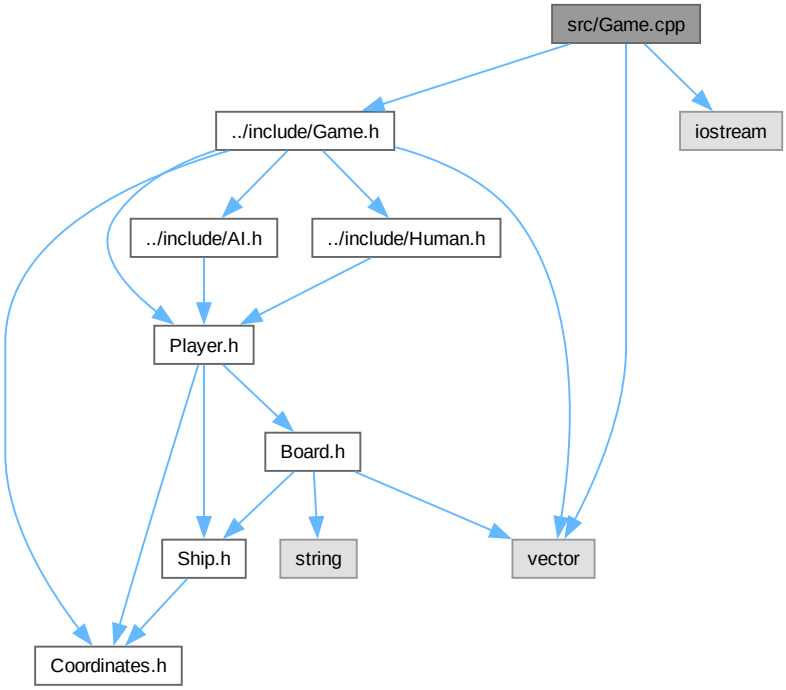
8.18 src/Game.cpp File Reference

Implementation of Game class.

```
#include "../include/Game.h"
#include <iostream>
```

```
#include <vector>
```

Include dependency graph for Game.cpp:



8.18.1 Detailed Description

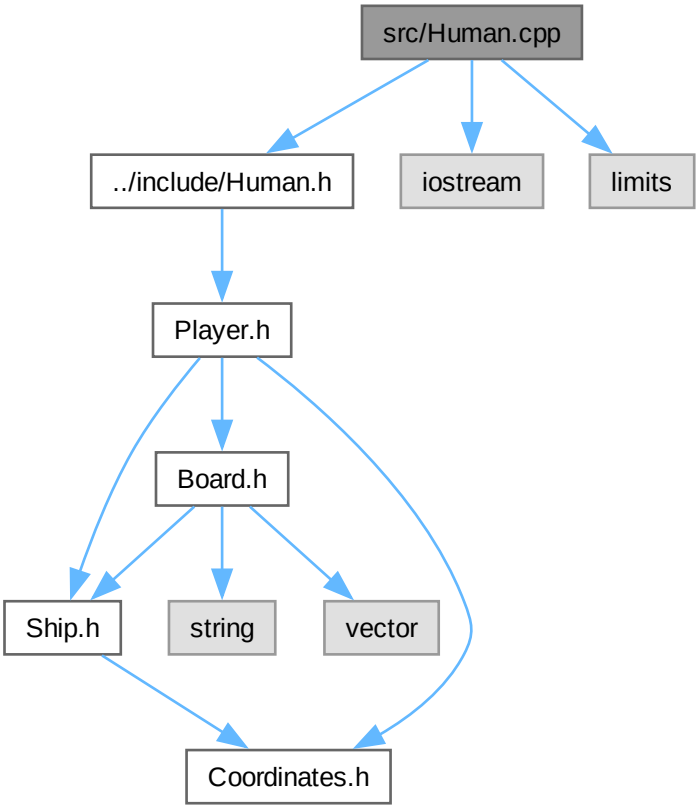
Implementation of Game class.

8.19 src/Human.cpp File Reference

Implements the Human class.

```
#include "../include/Human.h"
#include <iostream>
#include <limits>
```

Include dependency graph for Human.cpp:



8.19.1 Detailed Description

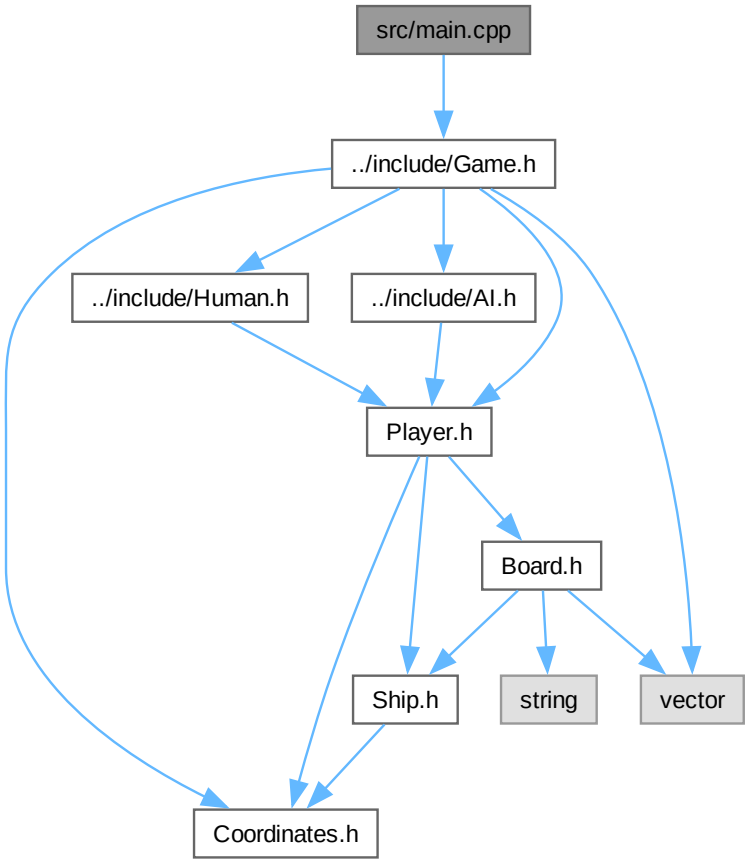
Implements the Human class.

8.20 src/main.cpp File Reference

main game file that runs the game.

```
#include "../include/Game.h"
```

Include dependency graph for main.cpp:



Functions

- int main ()

8.20.1 Detailed Description

main game file that runs the game.

8.20.2 Function Documentation

8.20.2.1 main()

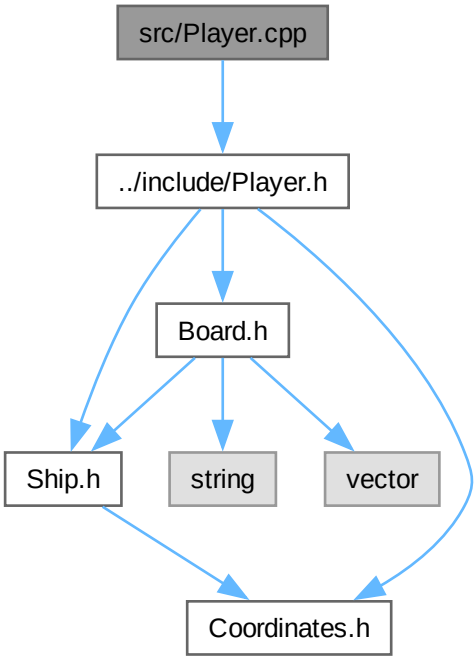
```
int main ()
```

8.21 src/Player.cpp File Reference

Implements the Player class, the abstract base for Human and AI players in a game of Battleship.

```
#include "../include/Player.h"
```

Include dependency graph for Player.cpp:



8.21.1 Detailed Description

Implements the Player class, the abstract base for Human and AI players in a game of Battleship.

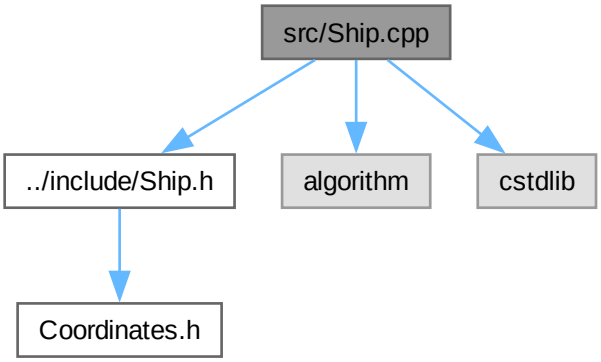
8.22 src/Ship.cpp File Reference

Implements the Ship class, which represents a ship in a game of Battleship.

```
#include "../include/Ship.h"
#include <algorithm>
```

```
#include <cstdlib>
```

Include dependency graph for Ship.cpp:



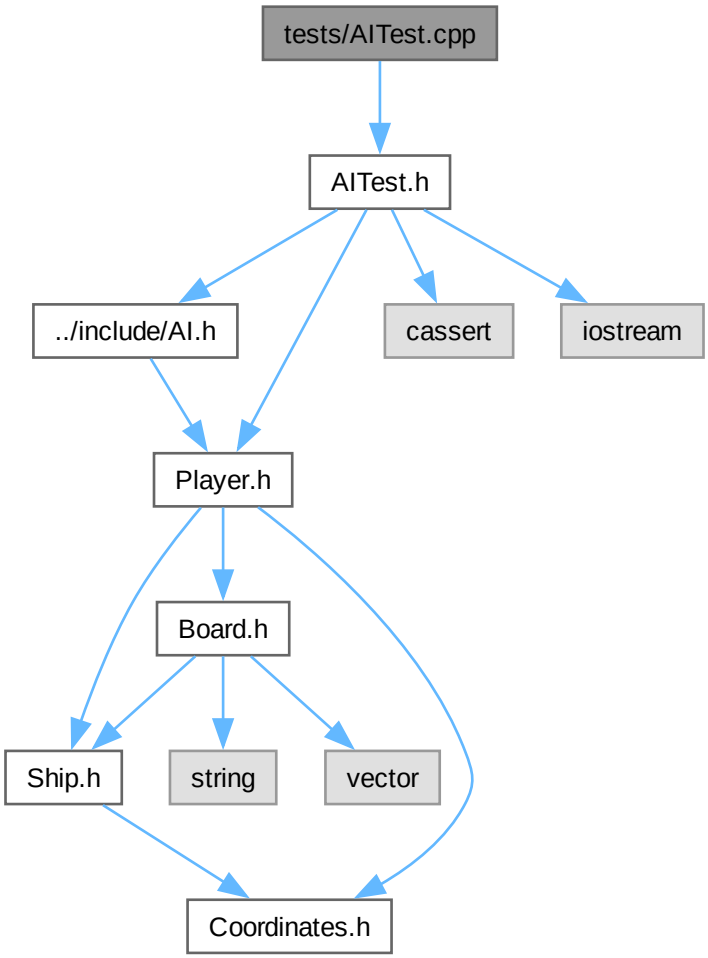
8.22.1 Detailed Description

Implements the Ship class, which represents a ship in a game of Battleship.

8.23 tests/AITest.cpp File Reference

```
#include "AITest.h"
```

Include dependency graph for AITest.cpp:



Functions

- void testAI ()

8.23.1 Function Documentation

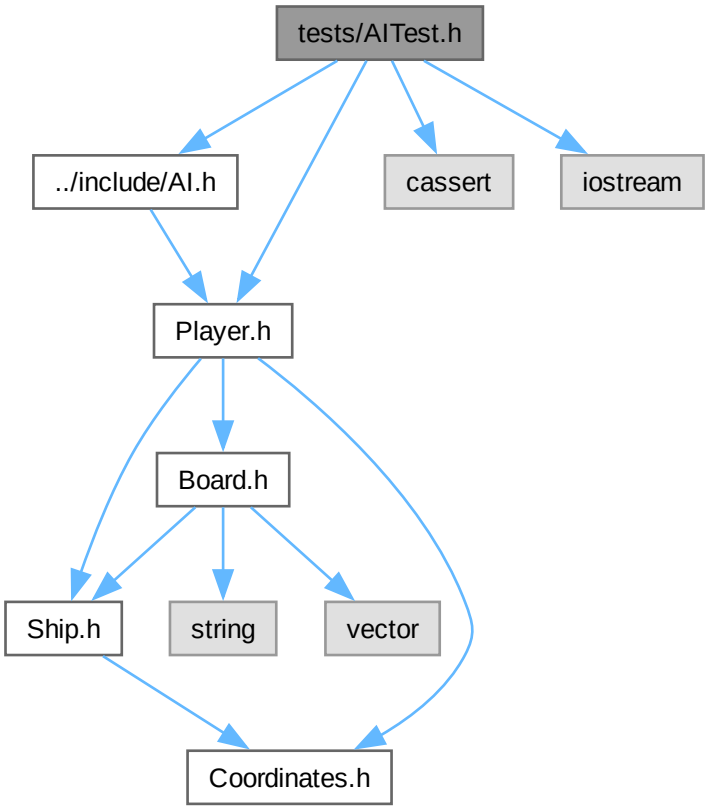
8.23.1.1 testAI()

```
void testAI ()
```

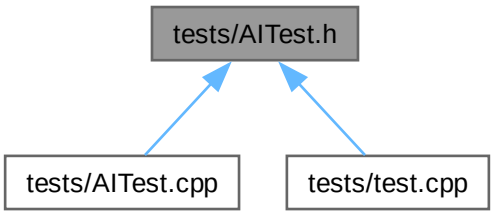
8.24 tests/AITest.h File Reference

```
#include "../include/AI.h"
#include "../include/Player.h"
#include <cassert>
#include <iostream>
```

Include dependency graph for AITest.h:



This graph shows which files directly or indirectly include this file:



Functions

- void testAI ()

8.24.1 Function Documentation

8.24.1.1 testAI()

```
void testAI ()
```

8.25 AITest.h

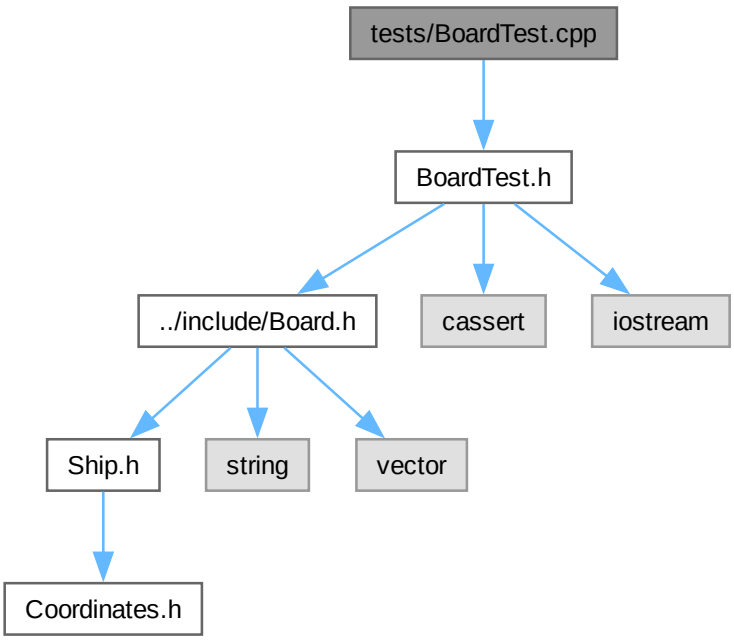
Go to the documentation of this file.

```
00001 #include "../include/AI.h"
00002 #include "../include/Player.h"
00003 #include <cassert>
00004 #include <iostream>
00005
00006 void testAI();
```

8.26 tests/BoardTest.cpp File Reference

```
#include "BoardTest.h"
```

Include dependency graph for BoardTest.cpp:



Functions

- void testBoard ()

8.26.1 Function Documentation

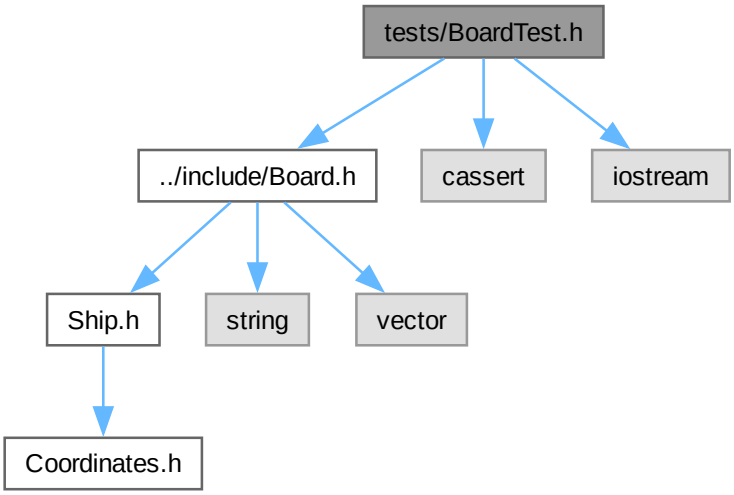
8.26.1.1 testBoard()

```
void testBoard ()
```

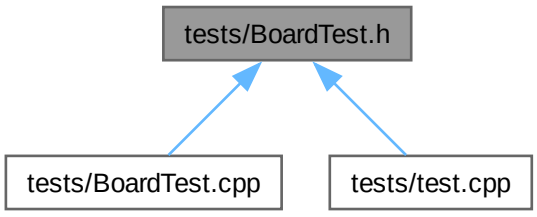
8.27 tests/BoardTest.h File Reference

```
#include "../include/Board.h"
#include <cassert>
#include <iostream>
```

Include dependency graph for BoardTest.h:



This graph shows which files directly or indirectly include this file:



Functions

- void testBoard ()

8.27.1 Function Documentation

8.27.1.1 testBoard()

```
void testBoard ()
```

8.28 BoardTest.h

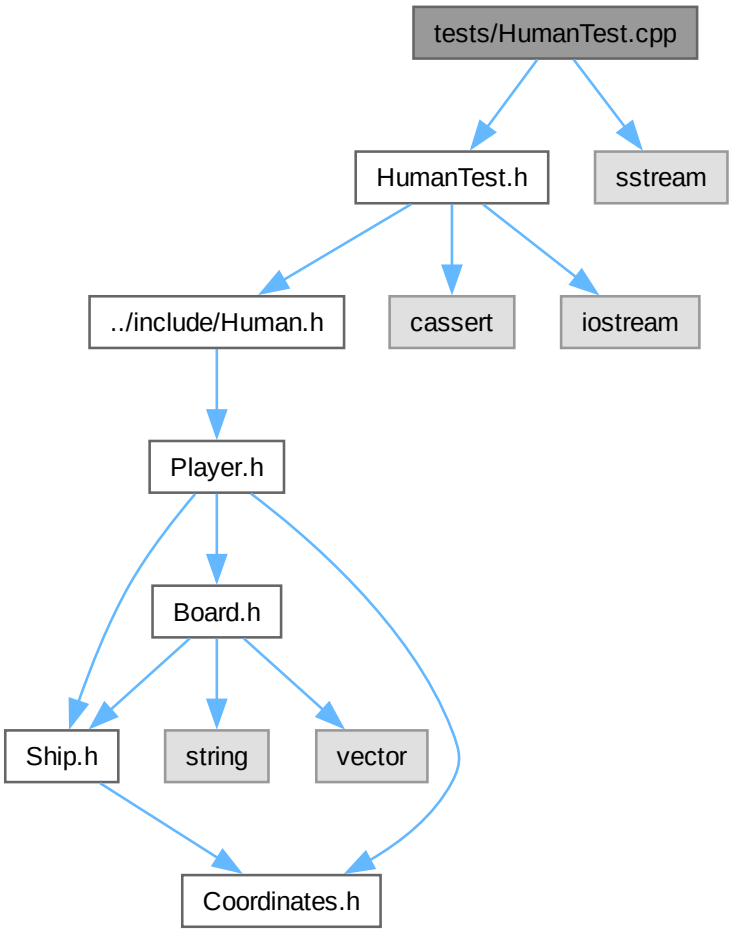
Go to the documentation of this file.

```
00001 #include "../include/Board.h"
00002 #include <cassert>
00003 #include <iostream>
00004
00005 void testBoard();
```

8.29 tests/HumanTest.cpp File Reference

```
#include "HumanTest.h"
#include <sstream>
```

Include dependency graph for HumanTest.cpp:



Functions

- void testHuman ()

8.29.1 Function Documentation

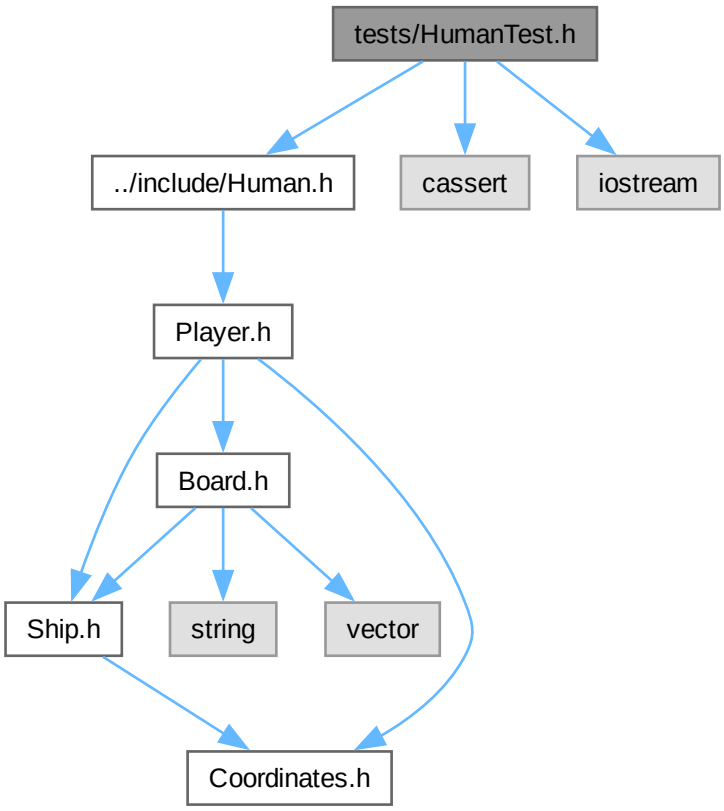
8.29.1.1 testHuman()

```
void testHuman ()
```

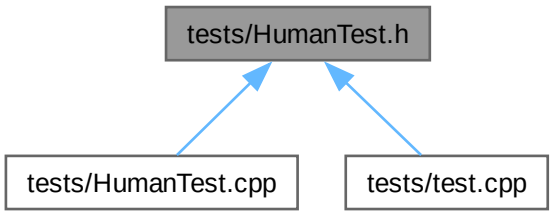
8.30 tests/HumanTest.h File Reference

```
#include "../include/Human.h"
#include <cassert>
#include <iostream>
```

Include dependency graph for HumanTest.h:



This graph shows which files directly or indirectly include this file:



Functions

- void testHuman ()

8.30.1 Function Documentation

8.30.1.1 testHuman()

```
void testHuman ()
```

8.31 HumanTest.h

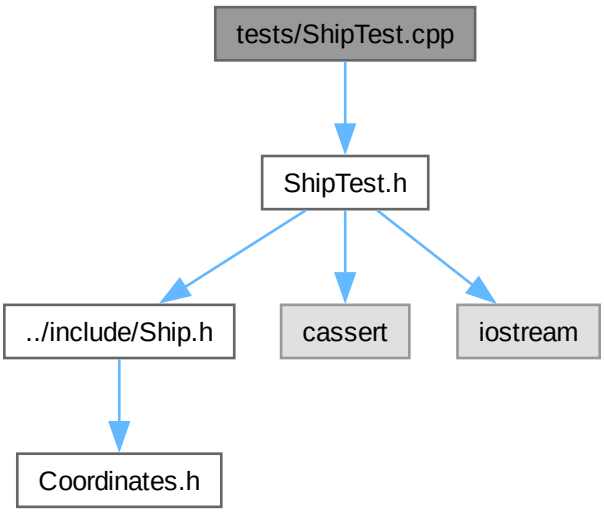
Go to the documentation of this file.

```
00001 #include "../include/Human.h"
00002 #include <cassert>
00003 #include <iostream>
00004
00005 void testHuman();
```

8.32 tests/ShipTest.cpp File Reference

```
#include "ShipTest.h"
```

Include dependency graph for ShipTest.cpp:



Functions

- void testShip ()

8.32.1 Function Documentation

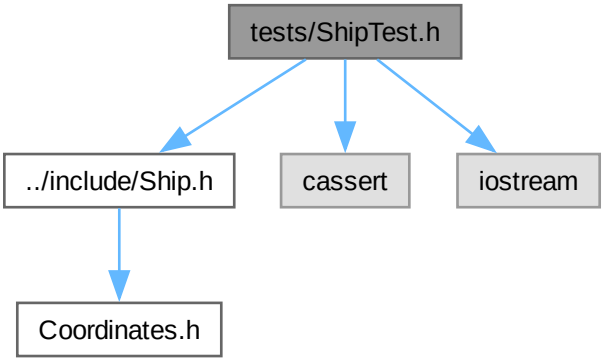
8.32.1.1 testShip()

```
void testShip ()
```

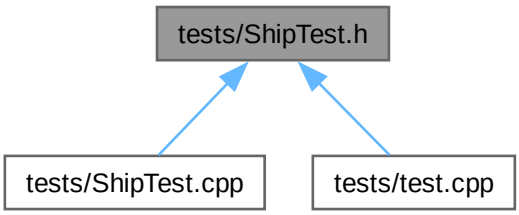
8.33 tests/ShipTest.h File Reference

```
#include "../include/Ship.h"
#include <cassert>
#include <iostream>
```

Include dependency graph for ShipTest.h:



This graph shows which files directly or indirectly include this file:



Functions

- void testShip ()

8.33.1 Function Documentation

8.33.1.1 testShip()

```
void testShip ()
```

8.34 ShipTest.h

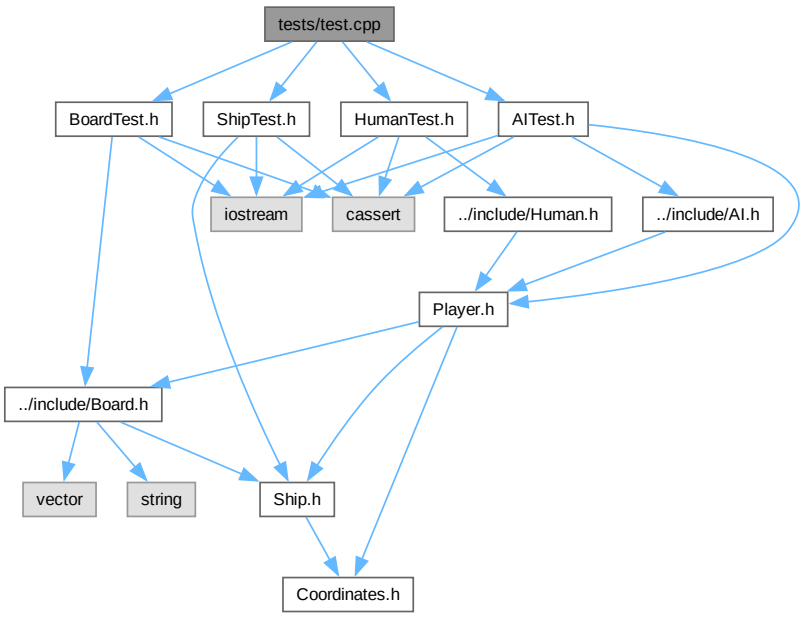
Go to the documentation of this file.

```
00001 #include "../include/Ship.h"
00002 #include <cassert>
00003 #include <iostream>
00004
00005 void testShip();
```

8.35 tests/test.cpp File Reference

```
#include "BoardTest.h"
#include "ShipTest.h"
#include "HumanTest.h"
#include "AITest.h"
```

Include dependency graph for test.cpp:



Functions

- int main ()

8.35.1 Function Documentation

8.35.1.1 main()

```
int main ()
```

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